CHANGE OF BASE SCHEME GIVES A 2-EQUIVALENCE BETWEEN ALGEBRAIC STACKS OVER S AND PAIRS (X', F) OVER S'

THE STACKS PROJECT AUTHORS

Editorial note. The source ends 'Hence the lemma is clear' after displaying both constructions; the identifications are carried as an explicit obligation.

Allowed standard prerequisites (editorial).

- Localization of stacks. Stacks, lemma localize-stacks (cross-chapter, named). Reduces the statement to preservation of the algebraic subcategories by constructions A and B.
- Constructions A and B. Stacks, section on localization (cross-chapter, structural). The two functors being compared.

Source reference. Original reference 04WV: focused-seeds/geometry/S9-alg-category-of-spaces-over-smaller-base-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04WT: focused-seeds/geometry/S9-alg-category-of-spaces-over-smaller-base-scheme/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let Sch_{fppf} be a big fppf site. Let $S \to S'$ be a morphism of this site. The constructions A and B of Stacks, Section 04WT above give isomorphisms of 2-categories*

$$\left\{\begin{array}{c}\text{2-category of algebraic}\\ \text{stacks } \mathcal{X} \text{ over } S\end{array}\right\} \leftrightarrow \left\{\begin{array}{c}\text{2-category of pairs } (\mathcal{X}', f) \text{ consisting of an}\\ \text{algebraic stack } \mathcal{X}' \text{ over } S' \text{ and a morphism}\\ f : \mathcal{X}' \to (Sch/S)_{fppf} \text{ of algebraic stacks over } S'\end{array}\right\}$$

Proof. The statement makes sense as the functor $j : (Sch/S)_{fppf} \to (Sch/S')_{fppf}$ is the localization functor associated to the object S/S' of $(Sch/S')_{fppf}$. By Stacks, Lemma 04WV the only thing to show is that the constructions A and B preserve the subcategories of algebraic stacks. For example, if $\mathcal{X} = [U/R]$ then construction A applied to $\mathcal{X}$ just produces $\mathcal{X}' = \mathcal{X}$. Conversely, if $\mathcal{X}' = [U'/R']$ the morphism p induces morphisms of algebraic spaces $U' \to S$ and $R' \to S$, and then $\mathcal{X} = [U'/R']$ but now viewed as a stack over S. Hence the lemma is clear. □

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.